An orbit is continuous falling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain a circular orbit with gravity and sideways motion.

SUCCESS CRITERIA - I can...

1. Describe orbit as free fall rather than absent gravity.
2. Distinguish tangent velocity from inward force.
3. Connect circular speed to orbital radius.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A projectile launches horizontally at **5 m/s** from height **20 m**.
1. Use g=**10** and no drag.
2. **Find flight time**, range and final vertical velocity.

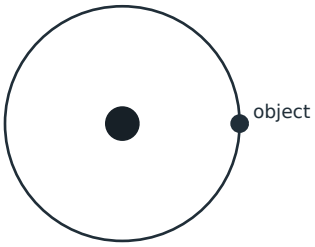
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $GM/r^2 = v^2/r$; $v = \sqrt{(GM/r)}$.
Circular orbit outside a spherical body of mass M; r is orbital radius. Neglect atmosphere and other bodies.

Worked reasoning: A small circular path used for practice has radius **4 m** and speed **2 m/s**. Its period is circumference/speed=**2**π×**4**/**2**=**4**π s. Gravity continuously changes the velocity direction.

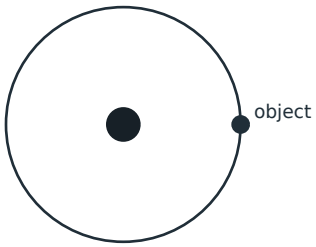
Your turn: Draw a quarter-orbit with tangent velocity arrows and inward gravitational arrows.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw the path of an** orbiting object and its velocity/force arrows.
2. **Explain how it can fall** without immediately hitting the surface.

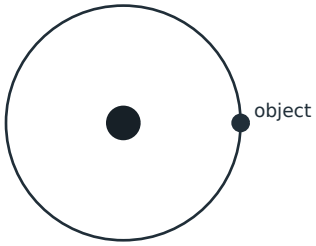


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

For a simulated planet GM=**400 m³/s²** and circular radius **16 m**, find orbital speed, period and acceleration. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: ① ② ③

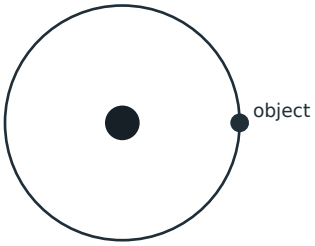
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Continue down each column.

05 Explain your reasoning

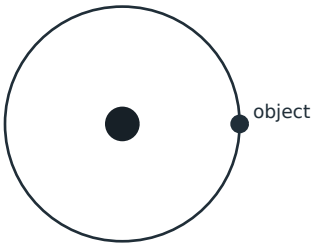
Why does a rocket not need to keep firing forward to remain in an ideal circular orbit?



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

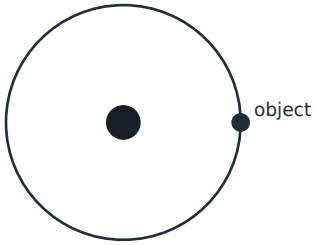
A **4 kg** object moves at **6 m/s** in an ideal circle under a force perpendicular to velocity. **Explain which parts of velocity** change and whether the force does work. **Label velocity** and inward acceleration on the circular-path diagram.



Marzano check - circle: 1 2 3

07 Exit check

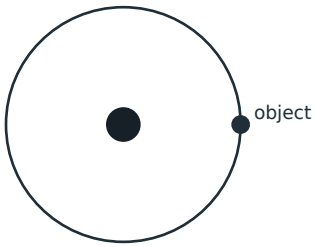
If circular orbital radius increases by factor **4** around same body, compare speed and period. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Around one source, a circular orbit radius increases ninefold.
1. **Compare speed**, period and gravitational acceleration without inserting a source mass.
2. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3